Meeting Cohort 10

Meeting Minutes

Date: 4 September

Time: 10:00

Chair group: Reinhard and Aatikah

Secretary group: Zoë and Jessie

Aobakwe Makole, Tokelo Mphahlele

Group 26G14 **Project 26P75**
Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

Accomplished This Week

- Started modelling power transfer and communication

Main Goals for Next Week

- Finish modelling

Abdullah Jamal, Faadhil Mohamed

Group 26G42 **Project 26P74**
AI-Based Traffic Prediction in Wireless Networks

Accomplished This Week

- Analysing short-term network traffic
- Started looking at existing solutions

Main Goals for Next Week

- Complete literature review on existing solutions and background

Ricco Kong, Thomas Liebenberg

Group 26G38 **Project 26P69**
Polarisation Through Disorder

Accomplished This Week

- Looking into existing solutions

- Start setup

Main Goals for Next Week

- Complete set up of system
- Begin collecting data

Jessie Cheng, Zoë da Silva

Group 26G36 **Project 26P71**
Engineering an empathetic control layer for LLMs via the Lövheim Cube

Accomplished This Week

- Looking into API and survey access

Main Goals for Next Week

- Finalize API and surveys
- Begin data collection

Reinhard Jansen van Vuuren, Aatikah Sheikh

Group 26G21 **Project 26P83**
Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

Accomplished This Week

- Began setup and characterisation
- Research

Main Goals for Next Week

- MZM characterisation
- Optical setup

Bilaal Essa, Mmatlou Marakalala

Group 26G47 **Project 26P21**
Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

Accomplished This Week

- Began working on circuit design

Main Goals for Next Week

- Finish circuit design
- Decide where ML will be used

Vukosi Khoza, Gift Vundla

Group 26G06 **Project 26P73**
AI-Assisted E-Learning Recommendation Platform

Accomplished This Week

- Generating more csv files

Main Goals for Next Week

- Start training models
- Extract info

General Cohort Comments

Actions or announcements for the cohort:

Next meeting:
Thursday 10 September, 10:00
Chair: Zoe and Jessie
Secretary: Thomas and Ricco

Meeting Cohort 10

Meeting Minutes

Date: 10 September

Time: 10:00 - 10:40

Chair group: Jessie and Zoë

Secretary group: Ricco and Thomas

Aobakwe Makole, Tokelo Mphahlele

Group 26G14 **Project 26P75**
Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

Accomplished This Week

- Used the Beer–Lambert law to determine the received power for the terahertz (THz) link
- Considered optical links for energy harvesting in a two-node (master/slave) system
- Implemented the directional water-filling algorithm for the master node only

Main Goals for Next Week

- Extend the directional water-filling algorithm to the slave node
- Obtain a proper estimate of the energy the master node will harvest over 8 hours
- Generate enough data to evaluate how well the algorithm performs, and establish its limitations

Additional Comments

None raised.

Abdullah Jamal, Faadhil Mohamed

Group 26G42 **Project 26P74**
AI-Based Traffic Prediction in Wireless Networks

Accomplished This Week

- Original dataset was removed from its source; a substitute dataset was found and is being evaluated for suitability
- Environment setup in progress
- Supervisor meeting held (administrative matters); a sample was sent to the supervisor

Main Goals for Next Week

- Complete the environment setup (expected today or tomorrow)
- Finalise the decision on whether the substitute dataset is suitable

Additional Comments

Removal of the original dataset has caused a delay while the substitute is evaluated.

Ricco Kong, Thomas Liebenberg

Group 26G38 **Project 26P69**
Polarisation Through Disorder

Accomplished This Week

- Constructed most of the bench setup, from the LG mode generation up to the scattering medium to produce speckle
- Reached the scattering stage and began testing different scattering mediums

Main Goals for Next Week

- Finish the setup
- Gather preliminary data and evaluate it
- Make adjustments to the setup based on the preliminary results

Additional Comments

Progress was delayed due to software issues.

Jessie Cheng, Zoë da Silva

Group 26G36 **Project 26P71**
Engineering an Empathetic Control Layer for LLMs via the Lövheim Cube

Accomplished This Week

- Completed the surveys and the API setup
- Loaded the survey questions
- Currently generating baseline results without adjusting the LLM parameters

Main Goals for Next Week

- Finish the baseline results
- Adjust the LLM parameters

Additional Comments

Issue: the LLM is not consistently following instructions; the group is working to figure out the cause.

Reinhard Jansen van Vuuren, Aatikah Sheikh

Group 26G21 **Project 26P83**
Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

Accomplished This Week

- No update recorded

Main Goals for Next Week

- Obtain an on-off keying (OOK) baseline
- Replace the arbitrary waveform generator

Additional Comments

None raised.

Bilaal Essa, Mmatlou Marakalala

Group 26G47 **Project 26P21**
Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

Accomplished This Week

- Completed the electronic design and paper design; started simulations and prototyping
- Integrated the subsystems and began making them communicate with each other
- Currently testing the filters and signal processing on the ESP32

Main Goals for Next Week

- Continue testing the system
- Implement a different network scheme for data transmission (e.g. LoRa or a mesh network)

Additional Comments

Issues with data transmission over long distances, and the transmitted data is inaccurate.

Vukosi Khoza, Gift Vundla

Group 26G06 **Project 26P73**
AI-Assisted E-Learning Recommendation Platform

Accomplished This Week

- Chose an embedding model (Hugging Face embeddings) to match patterns, converting user requests into vectors and matching them to content in the database
- Selected a separate model to respond to users (Gemini API), and began prompting the LLM
- Working on keyword-based information extraction and connecting related words using the vector model

Main Goals for Next Week

- Resolve the word-matching and case-handling issues
- Investigate a hybrid retrieval approach combining dense and sparse representations

Additional Comments

Issues with word matching and handling of cases. A possible mismatch was identified between using dense vectors for the database and sparse representations for requests; a hybrid model may address this.

General Cohort Comments

Issues affecting multiple groups:

- Software and environment setup delays (Groups 26G38 and 26G42)

- Data quality and availability concerns (dataset removal for 26G42; inaccurate data transmission for 26G47)

Actions or announcements for the cohort:

- All groups to come prepared with progress on the goals listed above for the next meeting

Next Meeting

Date: Friday, 18 September
Time: 09:30 – 10:30
Venue: EIE Seminar Room

Meeting Cohort 10

Meeting Minutes

Date:	18 September 2026
Time:	09:35
Chair group:	Group 26G38 (Ricco Kong, Thomas Liebenberg)
Secretary group:	Group 26G21

Aobakwe Makole, Tokelo Mphahlele

Group 26G14 **Project 26P75**
Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

Accomplished This Week

- Completed the derivation of solar and wave energy harvesting over a fixed time horizon (24-hour or hourly period).
- Reviewed the literature on the directional water-filling algorithm built last week and found that the first version had drifted from the project requirements.
- Began revising the water-filling algorithm to use Lagrangian optimisation.

Main Goals for Next Week

- Finish refining the directional water-filling algorithm using Lagrangian optimisation.
- Move on to the next stage of the project once the algorithm is finalised.

Additional Comments

The first version of the directional water-filling algorithm works but is not optimal, because it was implemented from the group's own interpretation. It is being reworked against the literature using a Lagrangian formulation.

Abdullah Jamal, Faadhil Mohamed

Group 26G42 **Project 26P74**
AI-Based Traffic Prediction in Wireless Networks

Accomplished This Week

- Met with supervisors, who said a comparison of existing one-hour-ahead traffic forecasting models would not add enough value to meet the GA requirements.
- Changed the project scope to a hybrid machine learning approach that is new to the group. The model will be chosen based on implementation cost and difficulty.
- Set up the development environment and validated the dataset.

Main Goals for Next Week

- Decide on the best way to run the simulation.
- Run the simulation within the next week.

- Settle on the hybrid model architecture based on cost and implementation difficulty.

Additional Comments

Scope changed after supervisor feedback. The group moved from a comparison of existing models to a new hybrid approach so the project adds enough value for the GAs.

Ricco Kong, Thomas Liebenberg

Group 26G38 **Project 26P69**
Polarisation Through Disorder

Accomplished This Week

- Finished the optical setup and took initial results; checked the speckle patterns, which looked good.
- Completed the first full data-gathering run, capturing 50 000 speckle images overnight.
- Chose a U-Net as the starting model, using a setup similar to the previous year's group.

Main Goals for Next Week

- Finalise the ML pipeline: data split, whether and how to downsample, and training configuration.
- Train the initial model (expected to finish around Monday), review the results and improve from there.
- Find a replacement diffuser that depolarises light more, or changes polarisation in a way that correlates with the speckle.

Additional Comments

The current diffuser changes polarisation very little, and the small change it does make is uniform. As a result the polarisation camera currently adds little over a standard intensity camera. A more depolarising diffuser should let the group get more value out of the polarisation data.

Jessie Cheng, Zoë da Silva

Group 26G36 **Project 26P71**
Engineering an empathetic control layer for LLMs via the Lövheim Cube

Accomplished This Week

- Completed the baseline and parameter-variation runs for the different surveys.
- Found gaps in emotion coverage and began adding three more surveys to fill them.

Main Goals for Next Week

- Carry out correlation analysis on the survey results.
- Add more surveys if further gaps are found.
- Explore a two-LLM interaction experiment: change one model's parameters and test whether the other picks up on the change in sentiment.

Additional Comments

The two-LLM sentiment experiment was suggested by Mitch and is an extension to the core scope.

Reinhard Jansen van Vuuren, Aatikah Sheikh

Group 26G21 **Project 26P83**
Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

Accomplished This Week

- Aligned all stages of the system end to end.
- Recalibrated the system; calibration drifts every few days and needs periodic adjustment.
- Identified that the waveform generator cannot meet requirements: it supports about 240 MHz (120 MHz for square waves), short of the gigahertz target, and the required voltage levels are outside its min/max output range.

Main Goals for Next Week

- Generate a stable high-frequency waveform using an FPGA with RF connections in place of the waveform generator.
- Transmit a bit stream through the system and receive it at the other end.
- Test the link over an outdoor free-space path (transmitting out of the window).

Additional Comments

The waveform generator's bandwidth and voltage-range limits are the main blocker for reaching gigahertz operation. The FPGA-based approach was being tested on the day of the meeting.

Bilaal Essa, Mmatlou Marakalala

Group 26G47 **Project 26P21**
Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

Accomplished This Week

- Completed the dashboard.
- Carried out early breadboard prototyping; the system is currently functional.
- Refined the work from last week while waiting for the LoRa components.

Main Goals for Next Week

- Implement the LoRa network once the antennas arrive.
- Continue refining and testing the breadboard prototype.

Additional Comments

LoRa implementation, planned for this week, was delayed because the ordered LoRa antennas have not arrived. Progress on the network depends on component delivery.

Vukosi Khoza, Gift Vundla

Group 26G06 **Project 26P73**
AI-Assisted E-Learning Recommendation Platform

Accomplished This Week

- Implemented the user interface.

- Carried out initial stress testing; the system handles the test cases correctly.
- Found that responses are too deterministic: answers follow a fixed template rather than being generated.

Main Goals for Next Week

- Integrate the Gemini API to generate responses, while keeping answers aligned with the course briefs.
- Carry out formal evaluation and testing of the system.
- Begin writing the report.

Additional Comments

Core implementation is functional. The main limitation is the templated, deterministic response style, which the Gemini integration aims to address. The focus is now shifting to evaluation so that report writing can start early.

General Cohort Comments

Issues affecting multiple groups:

- **Hardware and equipment constraints:** several groups are limited by equipment. 26G47 is waiting on LoRa antennas, 26G21's waveform generator cannot reach the required frequency or voltage levels, and 26G38's diffuser does not depolarise light enough to make use of the polarisation camera.
- **Scope refinement:** 26G42 (after supervisor feedback) and 26G14 (after a literature review) revised their approach to better match project requirements and GA expectations.
- **Shift to results:** most groups have working setups or implementations and are moving into data collection, model training, evaluation and reporting.

Actions or announcements for the cohort:

None recorded.

Meeting Cohort 10

Meeting Minutes

Date:	25 September 2026
Time:	09:00
Venue:	Wits EIE Reception, Jackson Room
Chair group:	Group 26G06
Secretary group:	Group 26G14

Aobakwe Makole, Tokelo Mphahlele

Group 26G14 **Project 26P75**
Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

Accomplished This Week

- Developed energy profiles for a master node that supplies energy to multiple slave nodes.
- Selected a lake-based location in Johannesburg for the solar energy profile, considering peak hours, moderate periods, sunrise and sunset. Harvesting periods are currently hard-coded.
- Investigated wave-energy harvesting in Cape Town, including the effect of wave height on electricity generation, and deep-sea thermal generation in Durban using temperature differences between surface and deeper water.
- Began implementing directional water-filling to allocate the available master-node energy between slave nodes.

Main Goals for Next Week

- Complete the machine-learning component for energy harvesting and integrate the different energy profiles.
- Continue development of the slave-node side and complete the directional water-filling implementation.
- Prepare a working simulation for the following week.

Additional Comments

Testing is planned as a simulation. Physical testing may be considered if the project is completed early.

Abdullah Jamal, Faadhil Mohamed

Group 26G42 **Project 26P74**
AI-Based Traffic Prediction in Wireless Networks

Accomplished This Week

- Discussed the project's novelty with the supervisor, who indicated that implementing and comparing existing models would not provide a sufficient contribution.
- Proposed developing a hybrid model or a new approach based on existing models.

- Completed validation of the dataset.

Main Goals for Next Week

- Finalise the new or hybrid model, compare it with existing models and run benchmark tests.
- Confirm any changes to the project title.
- Develop a GUI for the Open Day demonstration.

Additional Comments

The project needs an additional contribution beyond comparing existing models. Any resulting change to the project title still needs to be confirmed.

Ricco Kong, Thomas Liebenberg

Group 26G38 **Project 26P69**
Polarisation Through Disorder

Accomplished This Week

- Completed an initial dataset of approximately 50 000 images and trained baseline neural-network models for light-field reconstruction.
- Found that amplitude reconstruction is relatively straightforward, while phase reconstruction is complicated by the periodic equivalence of 0 and 2π.
- Implemented a loss function based on a vector representation of phase errors. Initial results indicate that it addresses the global phase issue.
- Investigated polarisation information and found that the current diffuser does not significantly affect polarisation, so using a polarisation camera would reduce image quality without a significant information benefit.

Main Goals for Next Week

- Test additional loss functions and different model architectures.
- Train models using the larger, more diverse dataset and investigate additional input combinations.
- Prioritise improvements to the current working system; investigate alternative diffusers if time permits.

Additional Comments

Following supervisor feedback, the group plans to remove polarisation from the current project scope. A conventional phase loss can penalise effectively correct predictions because 0 and 2π represent the same phase.

Jessie Cheng, Zoë da Silva

Group 26G36 **Project 26P71**
Engineering an empathetic control layer for LLMs via the Lövheim Cube

Accomplished This Week

- Completed the initial results and focused on analysing them during the week.
- Discussed an observed relationship between repetition penalty and serotonin.

- Received supervisor recommendations to use ANOVA, cross-correlation and PCA for statistical analysis.

Main Goals for Next Week

- Complete ANOVA, cross-correlation and PCA analyses.
- Generate improved visualisations and test additional intermediate parameter values.
- Begin preparing the project poster.

Additional Comments

The dataset has approximately eight dimensions, making visualisation difficult. Statistical analysis will help identify relationships and present the results more effectively.

Reinhard Jansen van Vuuren, Aatikah Sheikh

Group 26G21 **Project 26P83**
Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

Accomplished This Week

- Completed alignment of the final setup and began testing the system.
- Observed noisy, difficult-to-interpret outputs during initial data transmission and consulted the supervisor; the cause was not immediately identified.
- Found that the arbitrary waveform generator could not supply the required voltages at the required lower frequencies, despite its stated capabilities.
- Designed custom circuits to operate at the required lower frequencies.

Main Goals for Next Week

- Build the custom circuits and test their performance, effectiveness and efficiency.
- Continue evaluating how far the project can be developed.
- Begin preparing the Open Day poster.

Additional Comments

The unexpected transmission output and the waveform generator's limitations remain technical challenges. The custom circuits are intended to address operation at the required lower frequencies.

Bilaal Essa, Mmatlou Marakalala

Group 26G47 **Project 26P21**
Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

Accomplished This Week

- Collected components after availability-related delays and began assembling and soldering the system onto a Vero board.
- Completed 3D printing of the watch enclosure.
- Connected the main system and confirmed that it is working.

Main Goals for Next Week

- Continue soldering and hardware assembly.
- Resolve the charging configuration for the two batteries and investigate how they can be supplied with power in parallel.
- Continue testing the hardware system.

Additional Comments

The main remaining challenge is battery charging: the system has two batteries but only one charging component. The PCB machine is unavailable, so the group will continue using the Vero board for the remainder of the project.

Vukosi Khoza, Gift Vundla

Group 26G06 **Project 26P73**
AI-Assisted E-Learning Recommendation Platform

Accomplished This Week

- Discussed introducing novelty through an academic reviewer/advisor component being added to the existing system.
- Defined the component's purpose as assisting with uncertain eligibility or other decisions that require further review.
- Discussed a possible hybrid approach combining a newly developed method with an existing model; semantic and lexical search were mentioned as possible approaches to combine.

Main Goals for Next Week

- Obtain quantitative results and produce graphs and curves showing system performance.
- Compare the proposed system with existing advisor systems and evaluate whether the new component improves performance.
- Investigate whether the approach reduces false positives.

Additional Comments

The hybrid approach was discussed as a possibility rather than a finalised design. Evaluation will focus on whether the added reviewer/advisor component improves the existing system.

General Cohort Comments

Issues affecting multiple groups:

- **Novelty and scope refinement:** groups discussed additional contributions, hybrid approaches and changes to project scope in response to supervisor feedback.
- **Hardware and equipment constraints:** component availability, battery charging, the unavailable PCB machine and waveform-generator limitations are affecting progress.
- **Evaluation and presentation:** groups are moving into simulations, benchmarking, statistical analysis and preparation for Open Day demonstrations and posters.
- **Meeting participation:** there was concern about the limited time remaining before the project deadline and ensuring that every member/group has an opportunity to participate.

Actions or announcements for the cohort:

- Ensure that the presentation rotation gives every member/group an opportunity to participate in a meeting. An additional meeting may be arranged if required.
- **Next meeting:** 2 October 2026 at 09:00, Wits EIE Reception, Jackson Room.
- The next meeting will review progress against planned work, outstanding technical issues, Open Day preparation and final project requirements.

Meeting Cohort 10

Meeting Minutes

Date:	9 October 2026
Time:	09:00–10:00
Chair group:	Group 26G14 (Tokelo Mphahlele, Aobakwe Makole)
Secretary group:	Group 26G06 (Gift Vundla, Vukosi Khoza)

Aobakwe Makole, Tokelo Mphahlele

Group 26G14 **Project 26P75**
Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

Accomplished This Week

- Modelled the communication system and determined the reliable communication distance and signal-to-noise ratio.
- Calculated channel capacity and bit error rate for four modulation schemes: 4-QAM, 8-QAM, 16-QAM and 64-QAM.
- Prepared a simulation script and accompanying video following supervisor feedback.

Main Goals for Next Week

- Present the simulation video to the supervisor and address any resulting feedback.

Additional Comments
The group planned to show the simulation video to the supervisor immediately after the cohort meeting.

Vukosi Khoza, Gift Vundla

Group 26G06 **Project 26P73**
AI-Assisted E-Learning Recommendation Platform

Accomplished This Week

- Finalised the system results.
- Focused primarily on developing the project poster.
- Continued finalising the graphical user interface (GUI) for the chatbot.

Main Goals for Next Week

- Begin writing the project report.
- Continue refining the poster and chatbot GUI as needed.

Additional Comments
None.

Jessie Cheng, Zoë da Silva

Group 26G36 **Project 26P71**
Engineering an Empathetic Control Layer for LLMs via the Lövheim Cube

Accomplished This Week

- Completed further runs of the intermediate program.
- Conducted open-ended and antagonistic conversations to evaluate the large language model.
- Continued statistical analysis, results generation and poster preparation.

Main Goals for Next Week

- Complete the remaining statistical analysis and results.
- Finalise the poster and begin working on the project report.

Additional Comments
None.

Ricco Kong, Thomas Liebenberg

Group 26G38 **Project 26P69**
Polarisation Through Disorder

Accomplished This Week

- Performed scaling runs with different dataset sizes to assess the effect of training-set size on model performance; results suggest that the largest dataset has not yet reached a performance plateau.
- Repeated the strongest model configurations with different random seeds and observed inconsistent generalisation to previously unseen modes.
- Began implementing a rolling dataset of approximately 40,000 images, with newly captured images replacing the oldest, to increase data diversity without substantially increasing training time.
- Investigated image-resolution sensitivity by repeatedly downsampling and enlarging images from the original dataset. Training accuracy remained largely unaffected when the Laguerre–Gaussian (LG) mode patterns were clearly visible.
- Began repeating the resolution sweep on a new dataset with more propagated speckle patterns, where the underlying mode is less visible. Initial results show greater difficulty reconstructing the images.

Main Goals for Next Week

- Investigate why the best-performing configurations do not generalise consistently across random seeds.
- Continue evaluating the rolling-dataset approach.
- Complete the resolution sweep on the more strongly propagated speckle dataset and compare results with the original dataset.

Additional Comments
The resolution tests suggest that the model may be learning clearly visible mode patterns in the original dataset rather than robustly recovering patterns from heavily propagated speckle. The group is investigating this distinction.

Reinhard Jansen van Vuuren, Aatikah Sheikh

Group 26G21 **Project 26P83**
Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

Accomplished This Week

- Completed indoor data collection and turbulence measurements.
- Used the turbulence measurements to support bit-error-rate testing and comparisons between on–off keying and different polarisation shift keying states.
- Began outdoor alignment for an approximately 160 m optical link.
- Successfully aligned the outgoing beam onto the mirror at the far end of the link.

Main Goals for Next Week

- Complete the return-beam alignment from the distant mirror into the receiving optical setup.
- Continue outdoor testing once the link is aligned.

Additional Comments
The group aimed to complete the return-path alignment on the day of the meeting or over the weekend if necessary.

Bilaal Essa, Mmatlou Marakalala

Group 26G47 **Project 26P21**
Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

Accomplished This Week

- Continued arranging the circuit layout and 3D-printing the watch-face housing and related components.
- Conducted initial system testing.
- Worked on and substantially finalised the project poster.

Main Goals for Next Week

- Carry out the mock-mine tests, which were rescheduled to Tuesday, 13 October 2026.
- Meet with the supervisor to confirm whether changes to the poster are required.
- Make the remaining firmware updates and begin writing the project report.

Additional Comments
The group reported that the project was nearing completion, with final firmware changes and mock-mine testing still outstanding.

General Cohort Comments

Issues affecting multiple groups:

- Several groups are now preparing posters and transitioning towards writing their final reports.
- Outstanding work is concentrated on final validation, testing and presentation of results.

Actions or announcements for the cohort:

- No cohort-wide announcements were recorded in the supplied transcript.

Meeting Cohort 10

Meeting Minutes

Date:	2 October 2026
Time:	09:00–10:00
Chair group:	Group 26G47 (Bilaal Essa, Mmatlou Marakalala)
Secretary group:	Group 26G42 (Abdullah Jamal, Faadhil Mohamed)

Aobakwe Makole, Tokelo Mphahlele

Group 26G14 **Project 26P75**
Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

Accomplished This Week

- Completed the directional waterfilling algorithm.
- Began integrating the physical limitations of the components, and added a transmission schedule (e.g. 2, 5 or 10 minutes) to test how the algorithm responds to delayed transmission.
- Added the slave–master communication link using horn antennas at about 100 GHz, and started modelling the slave's energy use when active and idle to predict how long it lasts between harvests.

Main Goals for Next Week

- Determine the signal-to-noise ratio against reliable slave–master distance.
- Complete the communication link and obtain bit error rates.
- Check channel capacity to finish the model.

Additional Comments
In response to a question, the group clarified that the 100 figure refers to a 100 GHz carrier frequency, not 100 Gbps.

Abdullah Jamal, Faadhil Mohamed

Group 26G42 **Project 26P74**
AI-Based Traffic Prediction in Wireless Networks

Accomplished This Week

- Following supervisor feedback asking for a unique contribution, researched options and selected a hybrid forecasting model approach.
- Began building the simulation around the chosen hybrid approach.
- Structured the codebase modularly so each benchmark model can be validated independently without affecting the others.

Main Goals for Next Week

- Resolve the remaining code errors systematically.

- Validate the benchmark models, then implement the hybrid model.
- Allow one week for the poster, with a two-day buffer for final improvements.

Additional Comments
Progress is on track; only minor implementation issues remain.

Ricco Kong, Thomas Liebenberg

Group 26G38 **Project 26P69**
Polarisation Through Disorder

Accomplished This Week

- Tested different loss functions and model architectures.
- Achieved three-mode superpositions, producing a dataset just over double the previous size (more than three times the training data).
- Started training on the expanded dataset.

Main Goals for Next Week

- Let the current training run finish.
- Rerun the best configurations with different seeds to check consistency.
- Start the poster.

Additional Comments
Only modes of limited size pass through the system, and complex superpositions degrade, possibly due to SLM resolution. The lab network goes down overnight, delaying training. The loss functions trade off against each other with no net average gain.

Jessie Cheng, Zoë da Silva

Group 26G36 **Project 26P71**
Engineering an empathetic control layer for LLMs via the Lövheim Cube

Accomplished This Week

- Completed the statistical analysis.
- Obtained intermediate parameter values.
- Produced the visualisations.

Main Goals for Next Week

- Rerun the remaining graphs and statistics once the outstanding API calls complete.
- Work on the poster.
- Confirm with Mitch that he is satisfied with the project.

Additional Comments
API calls were timing out, apparently at the busiest times of day; this has improved. The project is otherwise largely complete.

Reinhard Jansen van Vuuren, Aatikah Sheikh

Group 26G21 **Project 26P83**
Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

Accomplished This Week

- Extended the project scope to use two intensity modulators.
- After self-splicing the bare fibre failed to collimate the beam, used most of the budget to have a technician repair the fibre.
- Aligned the system, which is now working well.

Main Goals for Next Week

- Collect all data for the poster and report.
- Identify the graphs that best present the findings.
- Carry out the investigation and comparisons.

Additional Comments
The fibre issue is resolved.

Bilaal Essa, Mmatlou Marakalala

Group 26G47 **Project 26P21**
Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

Accomplished This Week

- Received the outstanding components.
- Implemented most of the design and soldered the full circuit.
- Traced a failed MUX to an accidental short circuit and fixed it.

Main Goals for Next Week

- Fit the circuit into the helmet.
- Write the firmware and test the system.
- Complete the poster, aiming to finish by next week.

Additional Comments
None.

Vukosi Khoza, Gift Vundla

Group 26G06 **Project 26P73**
AI-Assisted E-Learning Recommendation Platform

Accomplished This Week

- Obtained the results of the system against the baselines.

Main Goals for Next Week

- Work on the poster.

Additional Comments
None.

General Cohort Comments

Issues affecting multiple groups:

- Unreliable shared infrastructure is delaying work: the lab network going down overnight (26G38) and API time-outs at busy times (26G36).
- Most groups are moving from implementation to data collection and poster preparation.

Actions or announcements for the cohort:

- The next meeting returns to the original group arrangement.
- The final cross-group meeting will be held on 14 October 2026, chaired by the group that has not yet chaired, so that every group has a turn. The chair group for that meeting is to be confirmed.
- No further cross-group meetings are scheduled after 14 October.

Investigation Cohort Meeting Minutes

Date:	16 October 2026
Recorded start:	07:05 (per transcript; timezone not specified)
Duration:	8 minutes 48 seconds
Source:	Microsoft Teams meeting transcript
Meeting facilitator:	Faadhil Mohamed (26G42)

1. Opening and Administration

Faadhil Mohamed facilitated the meeting and invited project groups to share their progress following Open Day. Bilaal Essa requested that speakers state their group numbers to support accurate transcription.

2. Group Progress Updates

2.1 Group 26G42 – Abdullah Jamal and Faadhil Mohamed

Project 26P74: AI-Based Traffic Prediction in Wireless Networks

- The group reported strong engagement during Open Day and promising results from its hybrid model, which slightly outperformed several alternatives.
- The supervisor was pleased with the findings and advised on how to present them in the technical report.
- The poster was completed in time for printing and display.
- **Next step:** Complete the final report and remaining project activities.

2.2 Group 26G21 – Reinhard Jansen van Vuuren and Aatikah Sheikh

Project 26P83: Performance Analysis of Polarisation Shift Keying in Free Space Optical Communication

- The group received useful feedback on its Open Day poster presentation.
- The group plans to improve how its technical work is communicated in the non-technical report.
- Sufficient data appears to have been collected for the technical report; the experimental setup will remain available in case further tests are needed.
- **Next step:** Draft the reports and address any gaps identified during writing.

2.3 Group 26G47 – Bilaal Essa and Mmatlou Marakalala

Project 26P21: Design and Investigation of an IoT-Based Real-Time Noise Monitoring System for Prevention of Occupational Hearing Loss in Mining Operations

- Additional testing since the previous meeting produced positive results.
- The group completed its poster and received positive feedback from its supervisors.
- Open Day attracted substantial interest, including from visitors associated with the mining industry and the School of Mining Engineering.
- Visitors expressed interest in the project's potential applications and suggested ways to improve the presentation of the work.

- **Next step:** Complete the technical report and prepare for the final presentations.

2.4 Group 26G14 – Aobakwe Makole and Tokelo Mphahlele

Project 26P75: Energy Harvesting and Cooperation in Underwater Terahertz (THz) Communications

- The group prepared materials for Open Day, including an explanatory video intended for non-technical visitors.
- The presentation was reported to have gone well. Parts of this update were unclear in the automatic transcript.
- **Next step:** Prepare the non-technical report and interview, and complete the final technical report.

2.5 Group 26G36 – Jessie Cheng and Zoë da Silva

Project 26P71: Engineering an Empathetic Control Layer for LLMs via the Lövheim Cube

- The group focused on its poster and received positive feedback during Open Day.
- Suggestions for future extensions of the project may be incorporated into the report.
- **Next step:** Focus on the final report.

2.6 Group 26G06 – Vukosi Khoza and Gift Vundla

Project 26P73: AI-Assisted E-Learning Recommendation Platform

- The group created an explanatory video for non-technical visitors and made minor improvements to its chatbot.
- Results and findings are available for reporting.
- **Next step:** Write the final report and incorporate the results.

2.7 Group 26G38 – Ricco Kong and Thomas Liebenberg

Project 26P69: Polarisation Through Disorder

- Reinhard Jansen van Vuuren relayed the group's update on its behalf.
- The group received positive feedback and intends to incorporate it into its reporting.
- **Next step:** Begin report preparation immediately or early the following week.

3. Shared Outcomes

Groups generally reported positive Open Day experiences and constructive feedback. Their remaining work centres on final technical and non-technical reports, incorporating feedback, and preparing for final assessments. Additional testing may be undertaken where report writing reveals gaps.

4. Action Items

Action	Responsibility	Timing
Finalise technical reports	All groups	Upcoming submission period
Refine non-technical communication and reports	Relevant groups	Before submission
Incorporate Open Day feedback	All groups	During report writing
Perform further testing if gaps emerge	Relevant groups	As needed
Prepare for final presentations/interviews	Relevant groups	Before assessments

5. Recognition and Closing

Faadhil Mohamed congratulated Reinhard Jansen van Vuuren and Aatikah Sheikh on winning the poster competition, wished the groups success with their reports, and closed the meeting.

Editorial note: Group memberships, group codes and project titles were cross-checked against the supplied 18 September 2026 LaTeX minutes. Progress statements reflect the 16 October 2026 transcript. The transcript's automatic transcription contains some unclear passages; uncertain details were not expanded into unsupported claims.